

4.3 Homogeneous Linear DEs with constant coefficients

Consider the 2nd order DE

$$a y'' + b y' + c y = 0 \quad (*)$$

where a, b, c are constants.

Likely candidates for solutions of $(*)$ are exponentials:

$$y = e^{mx}$$

$$\begin{aligned} y = e^{mx} &\rightarrow y' = m e^{mx} \\ &\rightarrow y'' = m^2 e^{mx} \end{aligned}$$

If we plug these into $(*)$ we get:

$$a [m^2 e^{mx}] + b [m e^{mx}] + c [e^{mx}] = 0$$

$$\Rightarrow e^{mx} (am^2 + bm + c) = 0.$$

Since $e^{mx} \neq 0$, then:

$$am^2 + bm + c = 0.$$

This is called the auxillary equation
(or characteristic equation) of the DE

$$ay'' + by' + cy = 0.$$

Let's consider three cases for the roots of the auxillary equation.

Case ① m_1 and m_2 are distinct real roots.

Case ② m is the only root.

Case ③ $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$ are complex conjugates roots.

Case ① m_1 and m_2 are distinct real roots

$$y_1 = e^{m_1 x} \text{ and } y_2 = e^{m_2 x}$$

are two solutions of the DE.

Are they LI?

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{m_1 x} & e^{m_2 x} \\ m_1 e^{m_1 x} & m_2 e^{m_2 x} \end{vmatrix}$$

$$W = (m_2 - m_1) e^{(m_1 + m_2)x}$$

$$W \neq 0 \text{ since } \underbrace{m_2 \neq m_1}_{\text{Distinct roots}}$$

$\Rightarrow$ they are LI.

$$\Rightarrow y = c_1 y_1 + c_2 y_2$$

General solution

$$\Rightarrow \boxed{y = c_1 e^{m_1 x} + c_2 e^{m_2 x}}$$

Case (2) m is the only root.

$y_1 = e^{mx}$ is a solution.

How do we find a pair of LI solutions?

In section 4.2 we learned that:

$$y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

Can be used to obtain another solution if a solution y_1 is known.

m is the unique root of

$$am^2 + bm + c = 0.$$

Roots : $m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

unique root $\rightarrow b^2 - 4ac = 0$

$\rightarrow \boxed{m = -b/2a}$

The DE expressed in standard form is

$$y'' + \underbrace{\frac{b}{a}}_{P(x)} y' + \frac{c}{a} y = 0$$

$y_1 = e^{mx}$ is a solution, where $m = -b/2a$.

$$y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

$$y_2 = e^{mx} \int \frac{e^{-\int \frac{b}{a} dx}}{[e^{mx}]^2} dx$$

$$y_2 = e^{mx} \int \frac{e^{-\frac{b}{a} x}}{e^{2mx}} dx$$

$$y_2 = e^{mx} \int \frac{e^{-\frac{b}{a} x}}{e^{-\frac{b}{a} x}} dx \quad \text{since: } 2m = -\frac{b}{a}$$

$$\Rightarrow \boxed{y_2 = x e^{mx}}$$

$y_1 = e^{mx}$ and $y_2 = x e^{mx}$ are
two L I solutions.

$$\Rightarrow y = c_1 e^{mx} + c_2 x e^{mx}$$

↑
General solution.

Quick review of Complex Numbers

$$i = \sqrt{-1}$$

$$\sqrt{-9} = 3i$$

$$Z = a + bi \quad \leftarrow \text{A complex number}$$

$$\overline{Z} = a - bi \quad \leftarrow \text{The } \underline{\text{conjugate}} \text{ of } Z$$

$$\begin{cases} a = \operatorname{Re}(Z) \leftarrow \text{Real part of } Z \\ b = \operatorname{Im}(Z) \leftarrow \text{Imaginary part of } Z \end{cases}$$

Complex roots of a polynomial with real coefficients always come in conjugate pairs.

For example : $m^2 + 2m + 5 = 0$

$$m = \frac{-2 \pm \sqrt{4 - 4(1)(5)}}{2}$$

$$m = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2}$$

$$m = -1 \pm 2i$$

The roots are the complex conjugates:

$$-1 + 2i \quad \text{and} \quad -1 - 2i.$$

Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Let's now look at case ③ for the roots of $am^2 + bm + c = 0$.

Case ③ $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$
are complex conjugate roots.

$$\text{Then: } \begin{cases} z_1 = e^{m_1 x} \\ z_2 = e^{m_2 x} \end{cases} \text{ and}$$

are complex solutions of the DE.

From the superposition principle,

$$y = K_1 Z_1 + K_2 Z_2$$

is a solution of the DE for any
Constants K_1 and K_2 .

$$y = K_1 e^{m_1 x} + K_2 e^{m_2 x}$$

$$y = K_1 e^{(\alpha + i\beta)x} + K_2 e^{(\alpha - i\beta)x}$$

$$y = K_1 e^{\alpha x} e^{i\beta x} + K_2 e^{\alpha x} e^{-i\beta x}$$

$$y = K_1 e^{\alpha x} \underbrace{(\cos \beta x + i \sin \beta x)}_{\text{Euler's Formula}} + K_2 e^{\alpha x} (\cos \beta x - i \sin \beta x)$$

$$\Rightarrow y = e^{\alpha x} \left((K_1 + K_2) \cos \beta x + i(K_1 - K_2) \sin \beta x \right)$$

$$\Rightarrow y = e^{\alpha x} \left(c_1 \cos \beta x + c_2 \sin \beta x \right)$$

where :
$$\begin{cases} c_1 = k_1 + k_2 \\ c_2 = i(k_1 - k_2) \end{cases}$$

By using the Wronskian, we can show that $y_1 = e^{\alpha x} \cos \beta x$ and $y_2 = e^{\alpha x} \sin \beta x$ are L.I.
$$\begin{cases} y_1 = \operatorname{Re}(e^{(\alpha+i\beta)x}) \\ y_2 = \operatorname{Im}(e^{(\alpha+i\beta)x}) \end{cases}$$

$$\Rightarrow y = c_1 y_1 + c_2 y_2$$

$$\Rightarrow \boxed{y = e^{\alpha x} \left[c_1 \cos \beta x + c_2 \sin \beta x \right]}$$

is the General solution.

To solve $ay'' + by' + cy = 0$,

We find the roots of: $am^2 + bm + c = 0$.

Case ① m_1 and m_2 are distinct real roots

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

Case ② m is the only root

$$y = c_1 e^{mx} + c_2 x e^{mx}$$

Case ③ $m = \alpha \pm i\beta$ are complex roots

$$y = e^{\alpha x} \left[c_1 \cos \beta x + c_2 \sin \beta x \right]$$

ex: ① Solve the IVP

$$\begin{cases} y'' - 5y' + 6y = 0 \\ y(0) = 3 \\ y'(0) = 4 \end{cases}$$

Step ① Solve the DE

We need to find the roots of $m^2 - 5m + 6 = 0$.

$$(m-2)(m-3) = 0 \rightarrow \underbrace{m=2, 3}_{\text{Distinct real roots}}$$

$$y = c_1 e^{2x} + c_2 e^{3x} \leftarrow \text{General solution}$$

Step ② Find c_1, c_2

$$y' = 2c_1 e^{2x} + 3c_2 e^{3x}$$

$$\begin{cases} y(0) = 3 \rightarrow c_1 + c_2 = 3 \\ y'(0) = 4 \rightarrow 2c_1 + 3c_2 = 4 \end{cases} \xrightarrow{\text{solve}} \begin{cases} c_1 = 5 \\ c_2 = -2 \end{cases}$$

The IVP solution is

$$y = 5e^{2x} - 2e^{3x}$$

ex: ② solve $y'' + 10y' + 25y = 0$.

Roots : $m^2 + 10m + 25 = 0$

$$(m + 5)^2 = 0$$

$$m = -5 \leftarrow \text{unique root}$$

$$\therefore y = c_1 e^{-5x} + c_2 x e^{-5x}$$

ex: ③ solve $y'' + 3y' = 0$.

Roots : $m^2 + 3m = 0$

$$m(m+3) = 0$$

$$m = 0, -3 \leftarrow \text{distinct real roots}$$

$$\therefore y = c_1 e^{0x} + c_2 e^{-3x}$$

$$y = c_1 + c_2 e^{-3x}$$

ex: ④ solve $y'' + y' + y = 0$.

Roots: $m^2 + m + 1 = 0$

$$m = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$m = \frac{-1 \pm \sqrt{-3}}{2}$$

$$m = \underbrace{-\frac{1}{2}}_{\alpha} \pm i \underbrace{\frac{\sqrt{3}}{2}}_{\beta}$$

We have two complex conjugate roots.

$$y = e^{\alpha x} \left[c_1 \cos \beta x + c_2 \sin \beta x \right]$$

$$y = e^{-\frac{x}{2}} \left[c_1 \cos\left(\frac{\sqrt{3}}{2}x\right) + c_2 \sin\left(\frac{\sqrt{3}}{2}x\right) \right]$$

ex: ⑤ Solve $y'' + 9y = 0$.

Roots : $m^2 + 9 = 0$

$$m^2 = -9$$

$$m = \pm 3i$$

$$m = \underset{\downarrow \alpha}{0} \pm \underset{\downarrow \beta}{3}i$$

$$y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

since $\alpha = 0$, then $e^{\alpha x} = e^{0x} = 1$

$$\Rightarrow \boxed{y = c_1 \cos(3x) + c_2 \sin(3x)}$$

In general: $y'' + \omega^2 y = 0$

is the DE for a Simple Harmonic Oscillator.

The general solution is

$$y = c_1 \cos(\omega x) + c_2 \sin(\omega x).$$

The solution can be expressed

as:

$$y = A \sin(\omega x + \phi).$$

More about this in chapter 5.

n^{th} order DE

Consider the DE

$$a_n y^{(n)} + \dots + a_2 y'' + a_1 y' + a_0 y = 0.$$

With constant coefficients:

$$a_n, \dots, a_2, a_1, a_0.$$

To solve it, we find the roots of:

$$a_n m^n + \dots + a_2 m^2 + a_1 m + a_0 = 0.$$

If there are n distinct real roots:

$$m_1, m_2, \dots, m_n$$

then:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$$

is the general solution.

ex: ⑥ solve $y''' - 4y'' + y' + 6y = 0$.

Roots: $m^3 - 4m^2 + m + 6 = 0$

$$(m+1)(m-2)(m-3) = 0$$

$$m = -1, 2, 3$$

$$\therefore y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{3x}$$

Repeated roots

If a root m is repeated k times, the general solution would include the k terms:

$$c_1 e^{mx} + c_2 x e^{mx} + \dots + c_k x^{k-1} e^{mx}.$$

ex: ⑦ solve $y''' - 6y'' + 12y' - 8y = 0$.

Roots: $m^3 - 6m^2 + 12m - 8 = 0$

$$(m - 2)^3 = 0$$

$$\Rightarrow m = 2 \quad (\text{Repeated 3 times})$$

$$\therefore y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}$$

ex: ⑧ solve the 4th order DE

$$y^{IV} + y''' - 7y'' - 13y' - 6y = 0.$$

Roots : $m^4 + m^3 - 7m^2 - 13m - 6 = 0$

$$(m-3)(m+2)(m+1)^2 = 0$$

$$\begin{cases} m_1 = 3 \\ m_2 = -2 \\ m_3 = -1 \end{cases} \quad [\text{Repeated twice}]$$

$$\Rightarrow y = c_1 e^{3x} + c_2 e^{-2x} + \underbrace{c_3 e^{-x} + c_4 x e^{-x}}_{\text{for } m_3 = -1}$$

Note : I used a Computer (Wolfram Alpha)
to factor the 4th degree Polynomial.